

Question Number	Answer	Additional guidance	Mark
8(a)	An explanation that makes reference to the following: • introduction of new (beaver) species increased the {species richness / biodiversity} (1) • reduction in (plant) biodiversity due to trees being cut down / reduction in (animal) biodiversity as {habitat / food} reduced (by the trees being cut down) (1) • (increase in biodiversity) due to {pond being created / new habitat / increased space (due to trees being cut down)} (1) • (increase in biodiversity) due to new food source (1)	Accept number of {species/ organisms} for biodiversity Ignore genetic diversity Accept reduction in animal biodiversity due to {competition with / predation from} beavers Accept plant or beaver as food source	(4)

Question Number	Answer
8 (b)	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material indicated as relevant. Additional content included in the response must be scientific and relevant. • genetic diversity identified as the variety of alleles in the population • only small numbers of introduced beavers could reduce the genetic diversity • loss of alleles from population • increased homozygosity • introducing more Norwegian beavers could increase genetic diversity • but could have same alleles so no effect on genetic diversity • introducing beavers from different European locations could increase genetic diversity • could be many different alleles in wider populations of European beavers • however the map shows many geographically isolated populations • these small populations could also have low genetic diversity • reproductive isolation could have occurred • preventing introduced beavers mating with Scottish beavers

			(6)
			Additional guidance
Level 0	0	No awardable content	
Level 1	1-2	Demonstrates isolated elements of biological knowledge and understanding to the given context with generalised comments made. Vague statements related to consequences are made with limited linkage to a range of scientific ideas, processes, techniques and procedures. The discussion will contain basic information with some attempt made to link knowledge and understanding to the given context.	Discussion of at least one solution and the impact on genetic diversity 1 solution = 1 mark 2 solutions = 2 marks
Level 2	3-4	Demonstrates adequate knowledge and understanding by selecting and applying some relevant biological facts/concepts. Consequences are discussed which are occasionally supported through linkage to a range of scientific ideas, processes, techniques and procedures. The discussion shows some linkages and lines of scientific reasoning with some structure.	Discussion of both solutions and their impact on genetic diversity plus consideration of effect of geographically isolated populations on genetic diversity 1 mark (accept description e.g. scattered populations) plus alleles 1 mark
Level 3	5-6	Demonstrates comprehensive knowledge and	all level 2 content

		understanding by selecting and applying relevant knowledge of biological facts/concepts. Consequences are discussed which are supported throughout by sustained linkage to a range of scientific ideas, processes, techniques or procedures. The discussion shows a well-developed and sustained line of scientific reasoning which is clear and logically structured.	plus reproductive isolation one mark plus inbreeding / heterozygosity index one mark
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Question Number	Answer	Additional guidance	Mark
8(c)	A calculation showing the following steps: • value for q (1) • value for p (1) • calculation of correct percentage for p^2 (1)	<u>Example of calculation</u> Accept values for either p or q: $\sqrt{0.09} / 0.3$ $(1 - 0.3) / 0.7$ $p^2 = 49\%$ Correct answer with no working scores full marks	(3)